Smart Electric Wheelchair with Regenerative Braking and Energy Optimization

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Abstract

Energy efficiency remains a critical challenge for assistive mobility devices. Battery-powered wheelchairs often suffer from limited range and frequent charging requirements. This project presents a smart electric wheelchair equipped with a regenerative braking system that recovers kinetic energy during deceleration and downhill motion. A microcontroller-based circuit manages energy flow, storing recovered power back into the battery to extend operating time. Integrated ultrasonic and slope sensors ensure safety by preventing collisions and regulating downhill speed. While the energy gain is modest (5–10%), the system significantly improves battery life and reduces maintenance. This work demonstrates the adaptation of electric vehicle regenerative technology for healthcare, aligning with national visions for efficient, inclusive innovation.

Introduction

Motivation:

Electric wheelchairs face critical range limitations and battery degradation while wasting braking energy as heat. By adapting regenerative technology from EVs, this project aims to enhance sustainability and accessibility.

Key Objectives:

- To recover energy during braking to recharge the battery.
- To enhance safety and comfort, and integrate sensors.
- To develop seamless drive-to-brake mode switching circuit.

Methodology

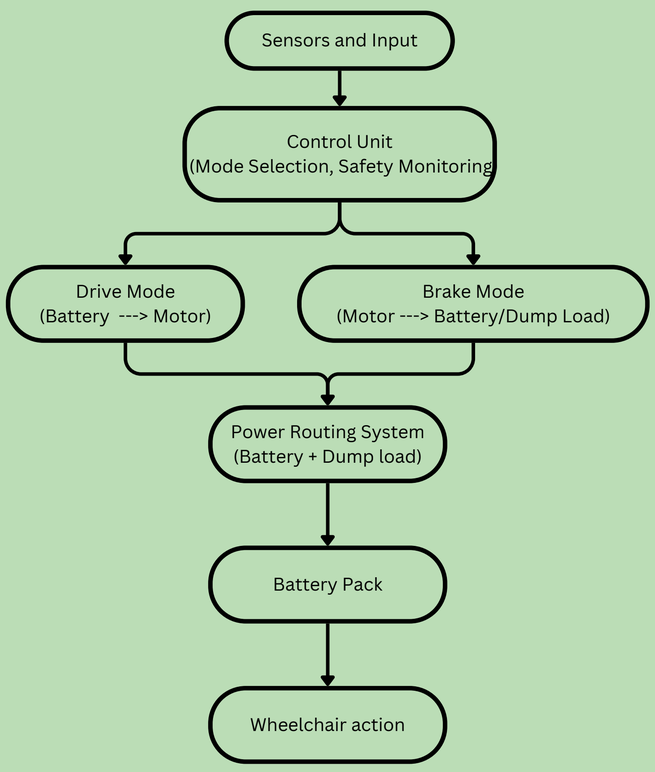


Figure 1: Methodology

Tool Used

ARDUINO

MATLAB

Expected Outcome

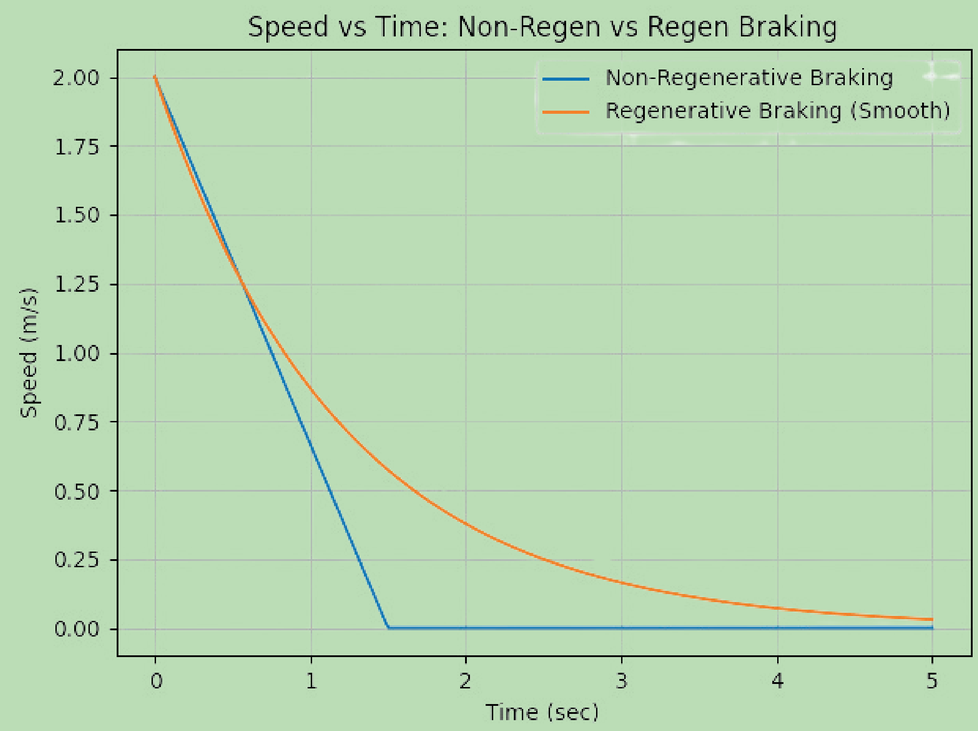


Figure 2: Comparison between non-regen and regen braking

Cost-Benefit Estimation

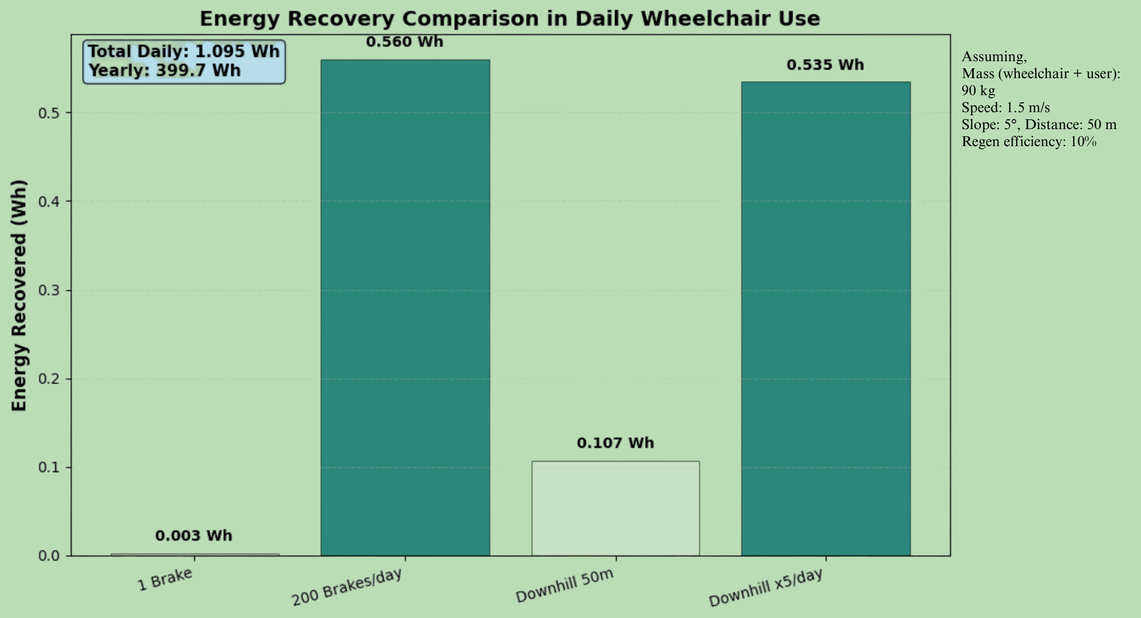


Figure 3: Energy recovery in normal and downhill ride

The one-time upgrade cost (~BDT 2.5-3.5k) yields small direct electricity savings but can extend battery life and reduce maintenance. Under realistic local battery prices, expected annual savings are ~BDT 400-1,500, giving a pragmatic payback of roughly 2-7 years. More importantly, the upgrade improves safety and comfort and demonstrates scalable micro-energy recovery— contributions that align with national energy-efficiency objectives.

Hardware Used

- Control Unit: Arduino/ESP32, Regen motor driver.
- Sensors: Ultrasonic, MPU6050, Voltage/Current.
- Power: 24V Battery, Capacitor Bank, MOSFETs/Diodes.

Conclusion

This project demonstrates that regenerative braking significantly improves efficiency and safety. By recovering kinetic energy, the system extends battery runtime while smart sensors ensure controlled downhill movement. This offers a practical, sustainable solution for assistive mobility, aligning with national goals for energy-conscious innovation.

References

1. Garade, U. et al. Regenerative Braking System in Electric Wheelchair, 2013.
2. Prasanth, B. et al. Maximizing Regenerative Braking Energy Harnessing in Electric Vehicles Using Machine Learning Techniques, 2023.